

Drone / Mobile Video to Walkable 3D Mesh World

Course: Software Engineering Lab (CS351)

Submission Mode: Google Classroom (Single PDF)

Submission Due Date: 02 October 2026

Target Hardware: 6GB Laptop GPU (RTX 3050/3060)

1. Software Requirements Specification (SRS)

IEEE-830 Format

1.1 Project Summary & Work Done

The objective of this project is to convert monocular video recorded from a drone or handheld smartphone into an interactive, physically walkable 3D environment inside a web browser. The core work done is summarized below:

- Video Ingestion & Keyframing:** Reads video (.mp4/.mov) and drops motion-blurred frames using Laplacian edge variance scoring to retain sharp photos with $\geq 70\%$ overlap.
- Camera Tracking (SfM):** Integrates COLMAP bundle adjustment to solve 6-DoF camera trajectories and sparse point landmarks with automated rescue ladders.
- 3D Radiance Training:** Trains 3D Gaussian Splats (3DGS) on a consumer 6GB laptop GPU under an active 5.8GB VRAM cap with checkpoint backups.
- Metric Scaling & Ground Fit:** Uses RANSAC ground plane fitting and physical flight speed/height anchors to scale arbitrary units into real meters (+Z upright).
- Physics Collision Meshing:** Builds 0.35m voxel boundary colliders and selects the optimal walkable ground surface (mesh vs. heightfield).
- Automated 11-Rule Quality Gate:** Tests ground orientation, player spawn clearance, and boundaries with zero tolerance for inverted floors.
- Autopilot Robot Walk Test:** Simulates a character walking 10 meters headless, verifying 0 fall events before approving scene release.
- Browser Delivery (WebGL):** Packages assets into a lightweight PlayCanvas HTML5 viewer running smoothly at 60 FPS on standard browsers.

1.2 Functional Requirements (System Feature Matrix)

Req ID	System Module	Functional Description	Pass Criterion
FR-101	Keyframe Extractor	Reads input video, analyzes frame edge sharpness, and discards motion-blurred frames.	Filters >80% of blur; retains $\geq 70\%$ inter-photo overlap.
FR-102	COLMAP Pose Solver	Matches SIFT feature points and solves 3D camera trajectory with rescue ladder fallbacks.	Registers $\geq 90\%$ of camera views successfully.
FR-103	Splat Trainer (3DGS)	Trains 350k 3D Gaussians for visual fidelity while capping GPU memory usage.	Peak VRAM ≤ 5.8 GB; saves partial checkpoints.
FR-104	Metric Ground Scaler	Fits ground plane via RANSAC and converts camera units to meters using physical anchors.	Scale error within $\pm 5\%$; upright Z-axis floor.
FR-105	Physics Mesh Builder	Constructs solid 0.35m voxel boundary colliders to block character clipping.	Character capsule cannot clip through walls or floors.
FR-106	11-Rule Safety Gate	Runs automated assertions on scale, floor orientation, and player spawn clearance.	100% pass on all 11 rules; emits receipt file.
FR-107	Autopilot Test Runner	Drives a character headless across the model for 10 meters, logging falls and coordinates.	Robot walks ≥ 10 m with exactly 0 falls.

2.1 DFD Level 0 — System Context Diagram

The Level 0 Context Diagram establishes the boundary of the system, showing how video data and user configurations enter the central engine and produce verified 3D assets for client exploration.



FIGURE 2.1: DFD LEVEL 0 — SYSTEM CONTEXT ARCHITECTURE

2.2 DFD Level 1 — Pipeline Data Flow

The Level 1 diagram traces data sequentially through all 6 pipeline processes and 5 local stores with clean, non-crossing orthogonal connections.

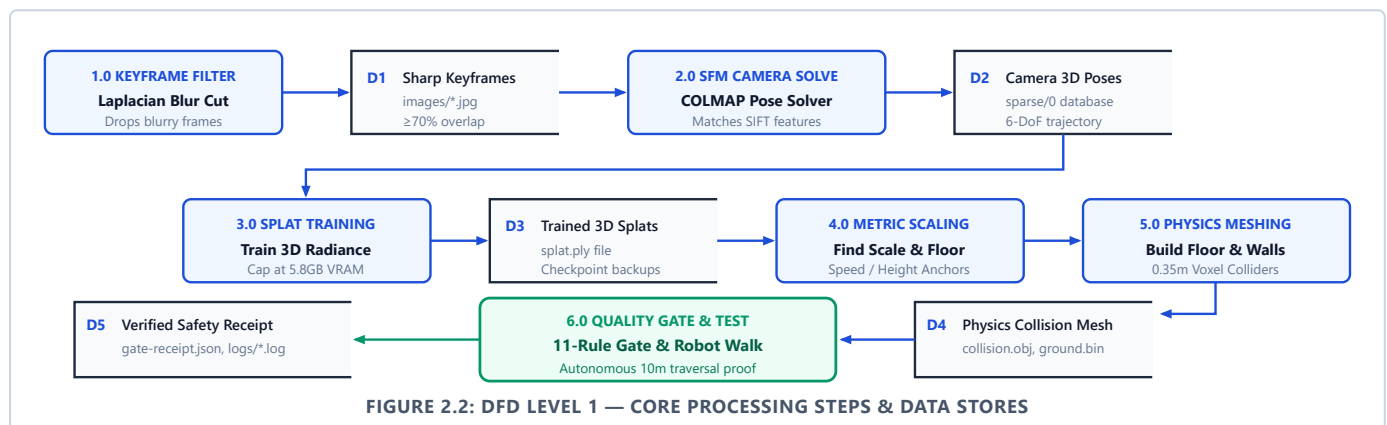


FIGURE 2.2: DFD LEVEL 1 — CORE PROCESSING STEPS & DATA STORES

2.3 DFD Level 2 — Detailed Floor Building & Safety Verification

The Level 2 diagram zooms into the ground extraction sub-process: fitting flat ground, generating candidate surfaces, picking the easiest walking route, and verifying the result with a robot character.

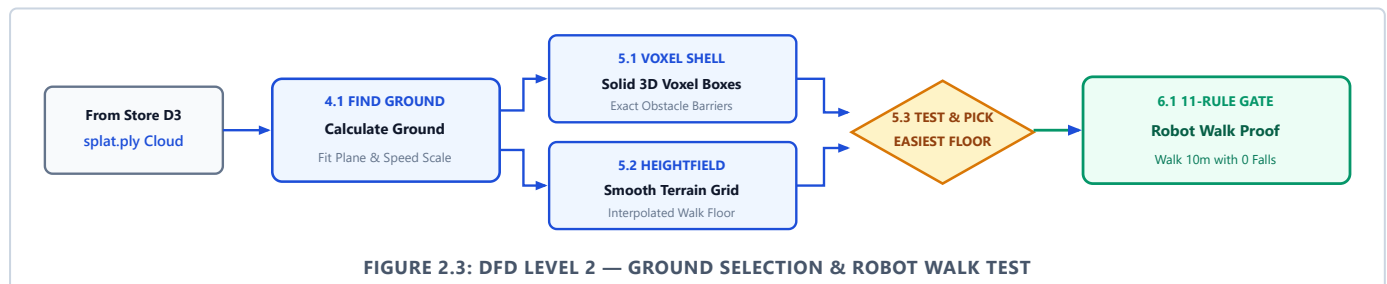
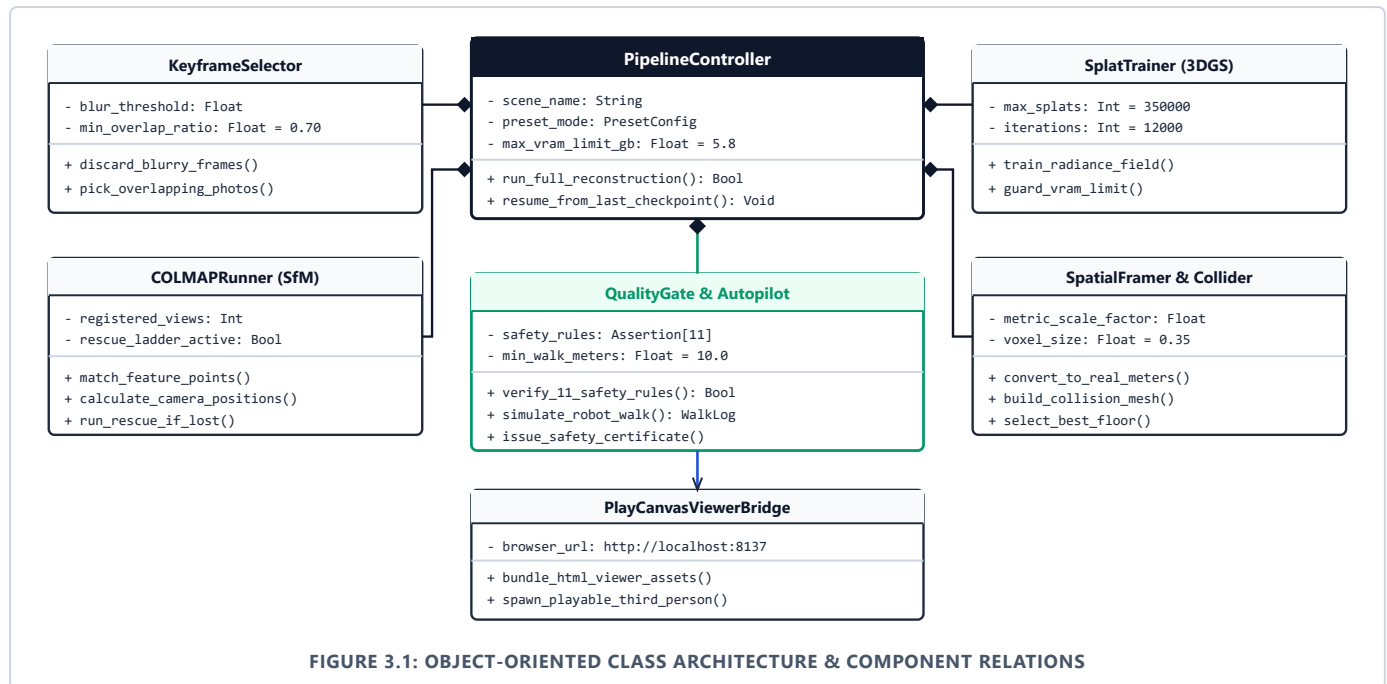


FIGURE 2.3: DFD LEVEL 2 — GROUND SELECTION & ROBOT WALK TEST

The Class Diagram models the software architecture. Classes are organized with clear attributes, private/public methods, and standard UML composition relationships (filled diamonds) and associations.



3.1 Class Responsibilities & Collaboration Dictionary

Class Name	Core Role & Implementation Responsibility	Direct Collaborators
PipelineController	Master orchestrator. Controls stage dependencies, checks resumability from logs, and catches hardware aborts.	Orchestrates all 5 worker classes.
KeyframeSelector	Pre-processing module. Computes Laplacian variance blur scores and decimates frames while maintaining $\geq 70\%$ visual overlap.	Outputs keyframe directory to COLMAPRunner.
COLMAPRunner	Structure-from-Motion engine. Matches feature points, runs bundle adjustment, and triggers rescue ladders if camera solving diverges.	Delivers camera poses to SplatTrainer.
SplatTrainer	Neural radiance field trainer (gsplat). Densifies and prunes 3D Gaussians while actively monitoring the 6GB VRAM memory ceiling.	Exports splat.ply to SpatialFramer.
SpatialFramer & Collider	Geometry engine. Fits the ground plane, applies speed/height anchors for metric scaling, and extrudes solid voxel collision shells.	Feeds collision mesh to QualityGate.
QualityGate & Autopilot	Verification test harness. Tests 11 assertions and drives a headless kinematic capsule character 10m to prove walkability.	Authorizes output release to PlayCanvasViewerBridge.

4.1 UML Use Case Diagram

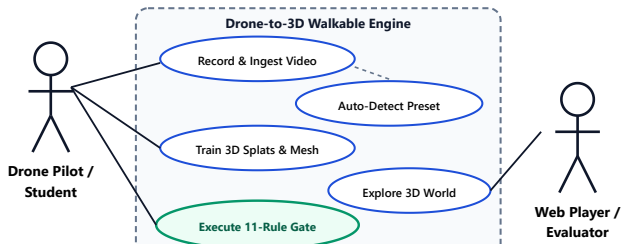


FIGURE 4.1: UML USE CASE INTERACTIONS

4.2 UML Sequence Diagram

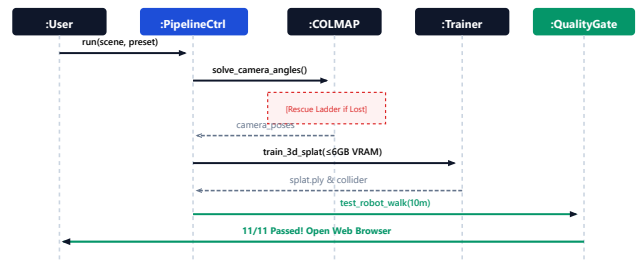


FIGURE 4.2: UML SEQUENCE FLOW & RESCUE

4.3 UML Activity & State Machine Flowchart

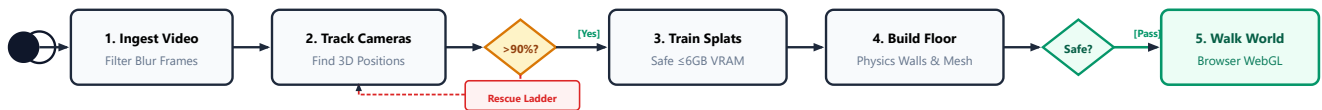


FIGURE 4.3: COMPLETE PIPELINE STATE MACHINE & EXECUTION PATH

4.4 Step-by-Step Pipeline Execution Summary

Stage	Internal Processing Actions	Observable User Output
1. Preflight	Scans video frames for motion blur and detects camera focal lengths.	Console logs: "Inspecting video... Presets tuned for room capture."
2. SfM Solve	Extracts feature points and tracks 3D camera angles. Automatic rescue ladder kicks in if solving diverges.	Console logs: "COLMAP registered 428/430 frames (99.5% success)."
3. 3D Splatting	Trains 350k Gaussian light ellipsoids, regularly checkpointing progress to prevent data loss.	Progress reaches 12,000 steps; VRAM consumption stays under 5.8 GB.
4. Floor & Walls	Fits ground plane to true meters and extrudes collision barriers for obstacles and boundaries.	Generates solid physics mesh (collision.obj) and floor elevations.
5. Robot Walk	Drives a headless avatar 10 meters through the space to ensure no falls through the floor occur.	Console logs: "Autopilot walk: 10.4m completed, 0 falls. Gate Passed!"
6. Web 3D	Launches an in-browser WebGL environment with keyboard navigation (WASD/arrows).	Automatically opens http://localhost:8137 at a smooth 60 FPS.